3.3.1.

1) $A=\begin{pmatrix}1&2\\3&2\end{pmatrix}$ $\quad C_A(x)=\det(A-x\,\mathcal{I}_n)$

$$C_A(x)=\begin{vmatrix}1-x&2\\3&2-x\end{vmatrix}=(1-x)(2-x)=2-x-2x+x^2-6=x^2-3x-4$$

$\Rightarrow Spec(A)=\{-1,4\}=(x+1)(x-4)$

$$\left.\begin{matrix}m_{-1}=1\ ,\ m_4=1\\ 1\le d_\lambda\le m_\lambda\end{matrix}\right\}\Rightarrow \begin{matrix}m_{-1}=d_{-1}=1\\ m_4=d_4=1\\ \sum m x=2\end{matrix}$$

$\Rightarrow A$ diag

Pt $\underline{\lambda=-1}$ $\quad \begin{pmatrix}2&2\\3&3\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}$

$\Rightarrow 2x_1+2x_2=0 \Rightarrow x_1=-x_2 \Rightarrow S_{-1}=\left\{\begin{pmatrix}-t\\t\end{pmatrix}\middle|\, t\in\mathbb{R}\right\}$

$\underline{\lambda=4}$: $\quad \begin{pmatrix}-3&2\\3&-2\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}$

$\Rightarrow -3x_1+2x_2=0 \Rightarrow x_1=\frac{2}{3}x_2$

$$S_4=\left\{\begin{pmatrix}\frac{2}{3}t\\t\end{pmatrix}\middle|\, t\in\mathbb{R}\right\}$$

$B_{-1}=\left\{\begin{pmatrix}-1\\1\end{pmatrix}\right\}$ $\quad B_4=\left\{\begin{pmatrix}-1\\1\end{pmatrix}\right\}$ $\quad P=\begin{pmatrix}-1&2\\1&3\end{pmatrix}$

$\Rightarrow P^{-1}AP=\begin{pmatrix}-1&0\\0&4\end{pmatrix}$

$A^n=?$ $\quad A^n=P\begin{pmatrix}-1&0\\0&4\end{pmatrix}^nP^{-1}$

$$A^n=-\frac{1}{5}\begin{pmatrix}-1&2\\1&3\end{pmatrix}\begin{pmatrix}(-1)^n&0\\0&4^n\end{pmatrix}\begin{pmatrix}3&-2\\-1&-1\end{pmatrix}$$

$$A^n=-\frac{1}{5}\begin{pmatrix}(-1)^{n+1}&2\cdot4^n\\(-1)^n&3\cdot4^n\end{pmatrix}\begin{pmatrix}3&-2\\-1&-1\end{pmatrix}=-\frac{1}{5}\begin{pmatrix}3(-1)^{n+1}-2\cdot4^n&-2(-1)^{n+1}-2\cdot4^n\\3(-1)^n-3\cdot4^n&(-1)^n\cdot(-2)-3\cdot4^n\end{pmatrix}$$

$$A^n = \begin{pmatrix} \frac{2\cdot 4^n + 3(-1)^n}{5} & \frac{2\cdot 4^n - 2(-1)^n}{5} \\ \frac{3\cdot 4^n - 3(-1)^n}{5} & \frac{3\cdot 4^n + 2(-1)^n}{5} \end{pmatrix}$$

b) $A = \begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix}$

$$C_A(x) = \det(xI - A) = \begin{vmatrix} x-2 & 4 \\ 1 & x+1 \end{vmatrix} = (x-2)(x+1) - 4 = x^2 + x - 2x - 2 - 4$$
$$= x^2 - x - 6 = (x-3)(x+2)$$

$\Rightarrow$ $Spec(A) = \{-2, 3\}$

$$\left.\begin{array}{l} m_{-2} = 1,\ m_3 = 1 \\ 1 \leq d_2 \leq m_{-2} \Rightarrow d_{-2} = 1 \\ 1 \leq d_3 \leq m_3 \Rightarrow d_3 = 1 \end{array}\right\} \Rightarrow A \text{ diagonalizabila}$$

c) $A = \begin{bmatrix} 7 & 0 & -4 \\ 0 & 5 & 0 \\ 5 & 0 & -2 \end{bmatrix}$

$$C_A(x) = \begin{vmatrix} x-7 & 0 & 4 \\ 0 & x-5 & 0 \\ -5 & 0 & x+2 \end{vmatrix} = (x-7)(x-5)(x+2) + 20(x-5)$$
$$= (x-5)(x^2 - 5x + 6) = (x-5)(x-2)(x-3)$$

$Spec(A) = \{2, 3, 5\}$

$$\left.\begin{array}{l} m_2 = d_2 = 1 \\ m_3 = d_3 = 1 \\ m_5 = d_5 = 1 \end{array}\right\} \Rightarrow A \text{ diag}$$

$$A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ -1 & 0 & 2 \end{pmatrix}$$

$$C_A(x) = \det(xI_3 - A) = \begin{vmatrix} x-1 & -1 & 0 \\ 1 & x & -1 \\ 1 & 0 & x-2 \end{vmatrix} = (x-1)x(x-2) + 1 +$$

$$+ x - 2 = (x-1)(x^2-2x+1) = (x-1)(x-1)^2 = (x-1)^3$$

$\operatorname{Spec} A = \{1\}$

multiplicitatea algebrică este n-rang

$m_1 = 3$

$$\operatorname{rang}\begin{pmatrix} 2 & -1 & 0 \\ 1 & 3 & -1 \\ 1 & 0 & 1 \end{pmatrix} = 2$$

$d_1 = 1$

$\Rightarrow$ A nu e diagonalizabilă

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 1 & -1 \\ 1 & 0 & -1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} -x_2 = 0 \\ x_1 + x_2 - x_3 = 0 \\ x_1 - x_3 = 0 \end{cases}, \quad x_3 = t$$

$x_1 = x_3 = t$

$x_2 = 0$

$$S_1 = \left\{ \begin{pmatrix} t \\ 0 \\ t \end{pmatrix} \,\middle|\, t \in \mathbb{R} \right\}$$

$$\dim \operatorname{Ker} f + \underbrace{\dim \operatorname{Im}}_{\text{rang}} = n$$

$$B_1 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 1 & -1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 1 & -1 \\ 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & -1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix}$$

$$\left\{ \begin{pmatrix} \alpha \\ \beta \\ \alpha+\beta \end{pmatrix} \middle| \ \alpha, \beta \in \mathbb{R} \right\}$$

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

- $(xI_3 - A)^3 = 0$, $(xI_3 - A)^3 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$\Rightarrow \ P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \Rightarrow P^{-1}AP$ triunghiulară

rez. la acest sistem

$P_1 : (xI_3 - A) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$ → prima coloană

dacă erau 2 dimensiuni luam 2 coloane

$P_2 : (xI_3 - A)^2 \begin{pmatrix} \\ \end{pmatrix} = 0$

3.4.1

a) $x_{k+2} = 3x_k + 2x_{k+1}$, $x_0 = 1$ $x_1 = 1$

$$V_k = \begin{bmatrix} x_k \\ x_{k+1} \end{bmatrix}$$

$$V_{k+1} = \begin{bmatrix} x_{k+1} \\ x_{k+2} \end{bmatrix} = \begin{bmatrix} x_{k+1} \\ 3x_k + 2x_{k+1} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_k \\ x_{k+1} \end{bmatrix} \Rightarrow A = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix} \qquad A^n = ?$$

$$C_x(A) = \det(xI - A) = \begin{vmatrix} x & -1 \\ -3 & x-2 \end{vmatrix} = x(x-2) - 3 = x^2 - 2x - 3 = (x+1)(x-3)$$

$Spec(A) = \{-1, 3\}$

$m_{-1} = m_3 = 1 \Rightarrow d_{-1} = d_3 = 1 \Rightarrow A$ diagonalizabilă

$\Rightarrow x_n = \alpha \cdot 3^n + \beta \cdot (-1)^n \Leftarrow$

$$V_{k+1} = A \cdot V_k = A^2 \cdot V_{k-1} = \ldots = A^{k+1} \cdot V_0 \Rightarrow \begin{bmatrix} x_{k+1} \\ x_{k+2} \end{bmatrix} = P \begin{pmatrix} (-1)^{k+1} & 0 \\ 0 & 3^{k+1} \end{pmatrix} \cdot P^{-1} \begin{pmatrix} x_0 \\ x_1 \end{pmatrix}$$

Mai punem condiț inițiale $x_0 = 1$, $x_0 = \alpha \cdot 3^0 + \beta(-1)^0 \Rightarrow \alpha + \beta = 1$

$x_1 = 1$, $x_1 = \alpha \cdot 3^1 + \beta(-1)^1 \Rightarrow 3\alpha - \beta = 1$ +

$4\alpha = 2 \Rightarrow \alpha = \frac{1}{2} \Rightarrow \beta = \frac{1}{2}$

$\Rightarrow x_n = \frac{1}{2} \cdot 3^n + \frac{1}{2}(-1)^n$

3.4.2 $x_{k+3} = 6x_{k+2} - 11x_{k+1} + 6x_k$, $x_0 = 1$, $x_1 = 0$, $x_2 = 1$

a) $$V_k = \begin{pmatrix} x_k \\ x_{k+1} \\ x_{k+2} \end{pmatrix}$$

$$V_{k+1} = \begin{pmatrix} x_{k+1} \\ x_{k+2} \\ 6x_{k+2} - 11x_{k+1} - 6x_k \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{pmatrix}}_{A} \begin{pmatrix} x_k \\ x_{k+1} \\ x_{k+2} \end{pmatrix}$$

$$C_A(X) = \begin{vmatrix} x & -1 & 0 \\ 0 & x & -1 \\ -6 & 11 & x-6 \end{vmatrix} = x^2(x-6) - 6 + 11x = x^3 - 6x^2 + 11x - 6 =$$

sau căutăm div termenului liber

$$= x^3 - x^2 - 5x^2 + 5x + 6x - 6 = x^2(x-1) - 5x(x-1) + 6(x-1) =$$

$$= (x-1)(x-2)(x-3)$$

$$\mathrm{Spec}(A) = \{1, 2, 3\} \Rightarrow \left.\begin{matrix} m_1 = d_1 = 1 \\ m_2 = d_2 = 1 \\ m_3 = d_3 = 1 \end{matrix}\right| \Rightarrow A \text{ - diag.} \Rightarrow$$

$$\Rightarrow x_n = \alpha\, 1^n + \beta 2^n + \gamma 3^n = \alpha + \beta \cdot 2^n + \gamma \cdot 3^n$$

$$\begin{cases} x_0 = \alpha + \beta + \gamma = 1 \\ x_1 = \alpha + 2\beta + 3\gamma = 0 \\ x_2 = \alpha + \beta + 9\gamma = 1 \end{cases}$$

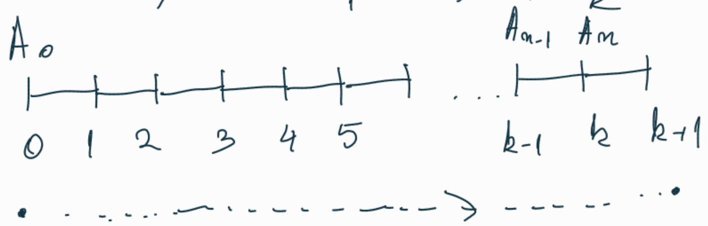
$$(2)-(1) \Rightarrow \begin{cases} \beta + 2\gamma = -1 \;/\cdot 4 \\ 3\beta + 8\gamma = 0 \end{cases} \Rightarrow \begin{cases} 4\beta + 8\gamma = -4 \\ 3\beta + 8\gamma = 0 \end{cases}$$

$$\beta = -4 \Rightarrow 2\gamma = 3 \Rightarrow \gamma = \frac{3}{2} \Rightarrow \alpha = \frac{7}{2}$$

$$x_n = \frac{7}{2} - 4 \cdot 2^n + \frac{3}{2} \cdot 3^n$$

<u>3.4.4</u>

pag 218 Știm: $S_1, S_2, \dots, S_k$



<u>Var.1</u> - ajung în k+1 pos

<u>Var 2</u> - ajung în (k-1) (pos dublu)

<u>Var 3</u> —||— $(k-1) + \underbrace{pas}_{\text{în } k} + pas$ => acest var. e inglobat în V_1

(în 2 pași simpli)

$$\Rightarrow S_{k+1} = S_k + S_{k-1}$$